

AYUSH KUMAR SINGH

+91 7002098294 • ayushkumarsingh683@gmail.com • National Capital Region (NCR), India
Portfolio • LinkedIn • GitHub

EDUCATION

ABES Institute of Technology (AKTU) <i>Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence)</i>	2024 – 2028 Ghaziabad, India Current SGPA: 7.13 / 10.0
Sampoorna Kendra Vidyalaya (CBSE) Senior Secondary (Class XII): 72.0% Secondary (Class X): 84.0%	Dibrugarh, Assam, India

TECHNICAL SKILLS

Frontend & Web: HTML5, CSS3, JavaScript (ES6+), Streamlit, Responsive UI Design
Backend & REST APIs: FastAPI, Python, Asynchronous Programming, RESTful Architecture, Pydantic V2, Uvicorn
Computer Vision & AI: OpenCV, YOLO11, Object Detection, Roboflow (Annotation & Augmentation), NumPy, Pandas, Scikit-Learn
Agentic Workflows & LLMs: LangGraph, LangChain, Multi-Agent Systems, Local LLMs (Ollama, Qwen 2.5), Moondream, Prompt Engineering
Databases & Cloud: PostgreSQL, MongoDB, MySQL Workbench, SQLAlchemy, Schema Normalization, Google Cloud Security
DevOps, Hardware & Tools: NVIDIA DGX A100 / Kubeflow, Docker, Git, GitHub, Linux/Bash, VS Code, Cursor

WORK EXPERIENCE

Global Infoventures (GI Infoventures) <i>AI & High-Performance Computing (HPC) Trainee</i> <i>Computer Vision Kubeflow NVIDIA DGX A100 Object Detection Roboflow Docker</i> - Configured and managed ML workflows on enterprise NVIDIA DGX A100 infrastructure via Kubeflow notebook servers. - Engineered computer vision datasets via Roboflow for annotation, preprocessing, and augmentation to train detection models. - Implemented real-time inference workflows using OpenCV, benchmarking performance across CPU and MIG GPU slices. - Containerized training workloads and runtime dependencies using Docker for isolated, reproducible execution.	Ghaziabad, India August 2026 – Present
KB QUBE <i>Machine Learning Intern</i> <i>Python Machine Learning Algorithms Data Preprocessing Scikit-Learn Predictive Modeling</i> - Completed practical ML assignments focused on supervised/unsupervised algorithms and structured dataset preprocessing. - Built automated data cleaning, missing-value imputation, and feature scaling pipelines using Python and Pandas. - Trained and validated predictive classification models using Scikit-Learn, evaluating precision, recall, and accuracy.	Dibrugarh, Assam, India 1 Month

KEY PROJECTS

Real-Time Edge Traffic Analytics & Vehicle Telemetry Pipeline <i>Engineered an end-to-end edge CV pipeline detecting, tracking, and classifying vehicles from 1080p surveillance feeds. Integrated YOLO11 Nano with ONNX Runtime and multi-threaded stream ingestion, sustaining stable 45+ FPS inference. Implemented ByteTrack for persistent tracking and built dual virtual tripwires to calculate vehicle velocity (km/h). Automated telemetry extraction logging track IDs, timestamps, speeds, and infraction flags into structured data sinks.</i>	Python YOLO11 ByteTrack OpenCV ONNX Roboflow
Autonomous Multi-Agent AI Travel & Itinerary Orchestrator <i>Architected a multi-agent system in LangGraph featuring 4 agents: Flight, Hotel, Itinerary, and Final Aggregator. Integrated live tool calling via Tavily Search and travel APIs for dynamic rate discovery, route mapping, and scheduling. Configured shared state memory (<code>TravelState</code>) with PostgreSQL persistence to preserve context across multi-turn sessions. Deployed Groq LPU (Llama 3) runtime for ultra-low latency response synthesis, cutting generation delay by 40%.</i>	Python LangGraph Groq (Llama 3) PostgreSQL Tavily API
Sovereign AI Workbench – Industrial Multi-Agent Inspection Platform <i>Engineered an on-premise industrial AI system leveraging local LLMs and vision models for private technical document analysis. Implemented a multi-agent pipeline using LangChain and Ollama to automate component extraction and risk scoring. Deployed multimodal vision models (Moondream / LLaVA) to detect mechanical anomalies in industrial diagrams. Exposed inference via asynchronous FastAPI endpoints with Pydantic payload validation and MongoDB persistence.</i>	Python LangChain Ollama (Qwen 2.5) Moondream FastAPI MongoDB

CERTIFICATIONS & PROFESSIONAL TRAINING

- | | |
|--|---|
| • AI/ML for Geodata Analytics – ISRO / IIRS | • Foundations of Prompt Engineering – AWS |
| • Trust & Security with Google Cloud – Google Cloud | • Data Analytics Job Simulation – Deloitte (Forage) |
| • Full Stack Development MasterClass – NoviTech R&D | • AI for Business Professionals – HP LIFE Foundation |